



Implications of the trends

The five trends that we have discussed so far cannot be considered in isolation, as together they represent significant challenges and opportunities for software developers and society going forwards.

The trends towards ubiquity and interconnectedness mean that while we can predict the way technology is evolving, it is still not clear what techniques will be used to successfully develop systems to exploit these trends. Despite rapidly evolving hardware, the ability to develop software that operates across several processors can pose issues, when this number is extrapolated to incorporate thousands of processors. These unimaginably large systems are often referred to as Global Computing.

The trends towards delegation and intelligence suggest that there is a growing need to develop computer systems that are capable of acting on our behalf, which means creating systems that are able to operate independently without direct supervision and that can represent our best interests.

The trends towards interconnection and distribution have been seen as a long standing issue and challenge within computer science and a significant amount of work and effort has gone into developing a range of distributed systems that can be created and deployed with greater ease. However, when we consider distributed systems in the context of also needing to represent our best interests this causes other issues. When we have delegated to a computer system to act in our best interests it will inevitably have to interact with other computer systems. These other systems are very likely to not have the same goals or interests. As such we need to then think about how we develop systems that have the ability to cooperate and reach agreements with one another.